CSE-4111: Parallel Processing and Distributed System

Question Bank

Compiled from University of Rajshahi Examination Papers (2019 - 2024)

Chapter 1: Fundamentals of Parallel & Distributed Processing

1. Why are we developing parallel systems instead of sequential systems?
   [Repeated 2 times: 2022-1a, 2020-1a]
2. Why do we need to write parallel programs (based on parallel algorithms)?
   [Repeated 2 times: 2022-1b, 2020-1b]
3. How can we exploit instruction-level parallelism and loop-level parallelism?
   [Repeated 2 times: 2022-1c, 2020-1c]
4. Parallelism can be achieved in a uniprocessor system using hardware and software technologies — what are they? Explain how parallelism is achieved using multiprogramming and timesharing methods.
   [1 time: 2021-2b/2c]
5. Define Parallel System and Distributed System. What are their key differences?
   [1 time: 2024-1a]
6. Define the terms: Parallel System, Distributed System, Multiprogramming, Timesharing, and Multi-processing.
   [1 time: 2023-1a]
7. What are the mechanisms of implementing parallel processing in a uni-processor system?
   [1 time: 2023-1b]
8. What do you mean by parallel processing?
   [1 time: 2019-1a]
9. Distinguish between parallel and distributed data processing.
   [1 time: 2023-1c]
10. Does parallel processing differ from multiprocessing? Explain your answer.
    [1 time: 2021-1c]
11. Why does increasing the number of processors not always improve performance?
    [1 time: 2024-1b]
12. 'Parallel computing is the simultaneous use of multiple computing resources to solve a computational problem.' Give real-life examples and write the main challenges for parallel computing.
    [1 time: 2021-1a]
13. What are the levels of parallel processing? Explain each with at least one example.
    [1 time: 2021-1b]
14. What challenges arise in parallel program design, and how can they be addressed?
    [1 time: 2023-2c]
15. Write some applications of parallel processing.
    [1 time: 2024-1c]

Chapter 2: Architecture Classification — Flynn's Taxonomy, CISC/RISC, VLIW, Superscalar

1. How does Flynn classify computers? / What do you mean by Flynn's Taxonomy? Explain how SIMD and MIMD computers work, with diagrams.
   [Repeated 2 times: 2022-2a, 2021-2a]
2. Differentiate between the characteristics of CISC and RISC processor architectures. / Define instruction set and briefly discuss CISC and RISC.
   [Repeated 2 times: 2022-2b, 2019-3a]
3. Draw and discuss the basic structure of an Array Computer.
   [1 time: 2024-2a]
4. Discuss the design space of major processor families.
   [1 time: 2024-2b]
5. Briefly explain the architecture of a superscalar processor.
   [1 time: 2024-2c]
6. Briefly explain the architecture, instruction format, and pipeline operations of a typical VLIW processor with a diagram.
   [1 time: 2022-2c]
7. Differentiate among scalar, superscalar, and vector processor.
   [1 time: 2019-3b]
8. Compare and contrast different types of parallel hardware architectures (SMP, MPP, GPU-based); give examples of applications.
   [1 time: 2023-2b]
9. How does parallel computing modify the traditional von Neumann model?
   [1 time: 2023-2a]
10. What is the bottleneck of the Von Neumann architecture?
    [1 time: 2020-2b]

Chapter 3: Pipelining

1. Classify pipeline processors according to the levels of processing.
   [Repeated 2 times: 2024-3a, 2022-4a]
2. Draw and explain the structure of a pipeline floating point adder.
   [Repeated 2 times: 2024-3b, 2022-4b]
3. Define linear, non-linear, synchronous, and asynchronous pipeline processor with a diagram. / Define linear and non-linear pipelines and discuss their differences.
   [Repeated 2 times: 2023-3a, 2021-8a]
4. Reservation table problem: state the forbidden/permissible latencies, calculate the collision vector, and draw the state transition diagram (numerical — table differs each year).
   [Repeated 2 times: 2021-8b, 2019-2b]
5. Explain the functional segmentation of a typical storage-to-register floating point instruction in the IBM 360/91.
   [1 time: 2022-4c]

6. Explain the working principle of a Carry Save Adder with an example of three 4-bit numbers.
   [1 time: 2023-4a]
7. Define horizontal processing, vertical processing, and vector looping in terms of pipelined vector processing.
   [1 time: 2023-4b]
8. A CPU of a modern computer is a good example of a linear pipeline. Explain.
   [1 time: 2021-4c]

Chapter 4: Speedup, Efficiency & Amdahl's Law

1. Define speedup, efficiency, and throughput.
   [Repeated 2 times: 2022-3a, 2021-4a]
2. How can you define the speedup factor, efficiency, and throughput of a k-stage pipeline processor over an equivalent non-pipelined processor?
   [1 time: 2019-2a]
3. A five-stage linear pipeline has clock period □=100ns and n=50 tasks. Find its throughput, efficiency, and speedup (numerical).
   [1 time: 2021-4b]
4. What is Amdahl's Law, and how does it limit the theoretical speedup of a parallel program? Illustrate with a numerical example.
   [1 time: 2024-3c]
5. Write down Amdahl's Law.
   [1 time: 2019-5c]
6. Compute the speed-up of each vector-processing method (horizontal/vertical/looping) with respect to a scalar processor for a given linear combination (numerical).
   [1 time: 2023-4c]

Chapter 5: Program Graphs, Data Dependence & Grain Packing

1. Discuss different types of data dependence. / Define different types of data dependence.
   [Repeated 2 times: 2019-1b, 2020-6b]
2. Draw the interconnection structure of a generalized multiprocessor system.
   [Repeated 2 times: 2024-4b, 2022-3c]
3. Define the three conditions of parallelism: data dependence, control dependence, and resource dependence.
   [1 time: 2023-8a]
4. 'Data dependency is a situation in which program statements refer to the data of a preceding statement.' What are the types of it? Explain with examples.
   [1 time: 2021-3a]
5. Consider the code fragment S1: Load R1,A; S2: Add R2,R1; S3: Move R1,R3; S4: Store B,R1 — find the dependencies between the statements.
   [1 time: 2021-3b]

6. Explain how I2 is resource dependent on I1 (I1: A=B+C, I2: D=G+H); draw the dependency graph.
   [1 time: 2021-3c]
7. Define pipenet. Show the program graph and the pipenet of a given equation E(I).
   [1 time: 2024-4a]
8. What is program graph transformation? Write the lemmas to convert a given program graph into an equivalent graph.
   [1 time: 2024-4c]
9. What is meant by resource conflict?
   [1 time: 2024-4d]
10. Define grain size and latency.
    [1 time: 2020-6a]
11. What are the steps of grain packing? Explain with an example.
    [1 time: 2020-6c]
12. Two 3×3 matrices A and B are given to compute the sum of the elements of the product matrix C=A×B; given inter-processor communication and node latencies, draw a fine-grain program graph and perform grain packing and parallel scheduling (numerical).
    [1 time: 2019-1c]
13. What is a data-flow computer? Write a simple C program and draw its dataflow graph.
    [1 time: 2019-1d]
14. What are the differences between static scheduling and dynamic scheduling?
    [1 time: 2023-8b]
15. Describe various interconnect architectures (Bus, Ring, Mesh, Hypercube, Crossbar) and their advantages/disadvantages.
    [1 time: 2023-8c]

Chapter 6: Memory Organization & Cache Coherence

1. Define interleaved memory organization. Compare high-order vs. low-order memory interleaving. / What is interleaved memory? Why is it beneficial for vector processors and parallel computing?
   [Repeated 2 times: 2022-3b, 2023-3b]
2. A system has 4-way interleaved memory with a given access/cycle time — calculate the effective memory bandwidth; then repeat for 8-way interleaving (numerical).
   [1 time: 2023-3c]
3. Which type of memory interleaving can tolerate a memory fault and which cannot? Explain.
   [1 time: 2019-3c]
4. If we fetch a cache line from main memory, where in the cache should it be placed?
   [1 time: 2020-2a]
5. What is cache coherency? With a suitable example, explain how cache becomes incoherent due to write-access of shared variables from multiple processors.
   [1 time: 2021-6a]

6. What is the cache coherence problem? Write a program to illustrate the cache coherence problem.
   [1 time: 2020-3c]
7. Discuss snooping cache coherence and directory-based cache coherence methods.
   [1 time: 2019-5b]
8. When two processors P1 and P2 have the same data element X and one modifies X to Y, explain the cache inconsistency problem for data sharing and process migration, in terms of write-through and write-back policies.
   [1 time: 2021-6b]

Chapter 7: Vector Processing

1. Explain the optimization of vector functions with an example. / Describe redundant expression elimination and constant folding at compile time.
   [Repeated 2 times: 2024-5c, 2019-8c]
2. What are the major principles for designing a vectorizer?
   [1 time: 2024-5a]
3. State different vector instruction types with an example.
   [1 time: 2024-5b]
4. Define vector and stride.
   [1 time: 2023-5a]
5. What do you mean by a vector task system? Explain with an example.
   [1 time: 2023-5b]
6. How could we enhance vector-processing capability?
   [1 time: 2023-5c]
7. What are the key characteristics of a vector processor?
   [1 time: 2020-3a]
8. Write some applications of a vector processor.
   [1 time: 2019-2c]
9. Define bisection width and bisection bandwidth.
   [1 time: 2019-5a]
10. Mention the desired features of a parallel language.
    [1 time: 2019-8a]
11. Convert the following sequential code to vector code: DO 20 I=1,N; A(I)=B(I-1); 20 B(I)=2*B(I).
    [1 time: 2019-8b]

Chapter 8: CUDA & GPU Computing

1. What is a CUDA kernel function?
   [1 time: 2019-4a]
2. What are the effects of launching CUDA kernel functions?
   [1 time: 2019-4b]

3. Write a CUDA kernel to perform addition of two vectors.
   [1 time: 2019-4c]
4. How can you set grid and thread block dimensions in CUDA programming?
   [1 time: 2019-4d]
5. Write short notes on GPU.
   [1 time: 2020-4c]

Chapter 9: Concurrency — Mutex, False Sharing, Threads & Synchronization

1. What is false sharing? Explain with an example.
   [Repeated 2 times: 2022-5a, 2019-7a]
2. What are the differences among mutex, busy-waiting, and semaphore?
   [Repeated 2 times: 2022-5b, 2019-7b]
3. Does false sharing generate an incorrect result? Explain the effects of false sharing.
   [1 time: 2020-3b]
4. Can multiple threads simultaneously insert new nodes into a linked list without any problem? If not, explain how, and write a code segment.
   [1 time: 2020-7a]
5. Can multiple threads simultaneously delete nodes of a linked list without any problem? If not, explain how, and write a code segment.
   [1 time: 2019-7c]
6. How can we synchronize threads?
   [1 time: 2020-7b]
7. What is mutex? Write a program using mutex.
   [1 time: 2020-7c]
8. What are the problems of busy-waiting?
   [1 time: 2020-8a]
9. Define critical section and race condition.
   [1 time: 2020-8b]
10. Given fibo[0]=fibo[1]=1 with a #pragma omp parallel for loop computing fibo[i]=fibo[i-1]+fibo[i-2], what will be the results if run with more than one thread?
    [1 time: 2020-8c]
11. Explain which function is thread safe and which is not.
    [1 time: 2020-2c]

Chapter 10: MPI Programming

1. What is the purpose of using the Allgather() function in MPI?
   [Repeated 2 times: 2022-5c, 2019-6c]
2. Why do we use MPI_Ssend instead of MPI_Send?
   [1 time: 2020-5a]

3. What are the limitations of MPI_Scatter and MPI_Gather functions?
   [1 time: 2020-2d]
4. How can the receiver find out the following values in MPI: the amount of data, the sender, and the tag of a message?
   [1 time: 2020-4a]
5. Why do we need the MPI_Reduce and MPI_Allreduce functions? Show the communication pattern of MPI_Allreduce.
   [1 time: 2020-4b]
6. What is MPI? Discuss the purposes and uses of MPI_Init, MPI_Finalize, MPI_Send, and MPI_Recv.
   [1 time: 2021-5b]
7. Write a program to read a vector and distribute it among processes using an MPI-provided function.
   [1 time: 2019-6a]
8. What are the abstract parallel programming models over hardware and memory architectures? Compare the shared memory model with the message passing model.
   [1 time: 2021-5a]

Chapter 11: Parallel Sorting Algorithms

1. Sort the numbers {15,11,9,16,3,14,8,7,4,6,12,10,5,2,13,1} using the parallel odd-even transposition sort algorithm.
   [Repeated 2 times: 2020-5b, 2019-6b]
2. Write a program in MPI to sort a set of numbers using the parallel odd-even transposition sort algorithm.
   [1 time: 2020-5c]
3. Given the array {12,14,6,8,10,16,2,4}, show all phases of sorting using the odd-even transposition sort algorithm.
   [1 time: 2021-7a]
4. Write a program using OpenMP to implement the odd-even transposition sort algorithm.
   [1 time: 2021-7b]

Chapter 12: Distributed Processing Fundamentals

1. Distinguish between local and remote distribution.
   [Repeated 2 times: 2024-6a, 2022-6a]
2. Write the main factors that trend to distributed processing.
   [Repeated 2 times: 2024-6b, 2022-6b]
3. Write/State the different categories of data distribution.
   [Repeated 2 times: 2022-6d, 2023-6b]
4. Briefly explain hierarchical distributed processing with an example.
   [1 time: 2024-6c]

5. What are the basic differences between function and system distribution?
   [1 time: 2022-6c]
6. Define software path length and teleprocessing.
   [1 time: 2023-6a]
7. What are the main reasons of function distribution?
   [1 time: 2023-6c]

Chapter 13: Distributed Database Systems

1. What is fragmentation? Differentiate between horizontal and vertical fragmentation with example; write their advantages. / Define data fragmentation in a Distributed Database and explain various types with example and advantages.
   [Repeated 3 times: 2024-7a, 2022-7b, 2023-7c]
2. How does Heterogeneous Distributed Databases differ from Homogeneous Distributed Databases? / Define Heterogeneous Distributed Databases and write its advantages. / Distinguish between homogeneous and heterogeneous distributed database.
   [Repeated 3 times: 2024-8c, 2022-8c, 2023-7b]
3. What is replication? Write some advantages and disadvantages of data replication.
   [Repeated 2 times: 2024-7b, 2022-8b]
4. What are the main reasons for failure of a distributed transaction?
   [Repeated 2 times: 2024-8b, 2022-7c]
5. Why do we use a commit protocol? Briefly explain the two-phase commit protocol.
   [Repeated 2 times: 2024-7c, 2022-8a]
6. Write the role of the transaction manager and transaction coordinator for a distributed system/database; how do they ensure consistency?
   [Repeated 2 times: 2024-8a, 2022-7a]
7. What is a Distributed Database and how does it differ from a centralized database?
   [1 time: 2023-7a]